

Prathik B S

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github.com/prathikbsgowda | Portfolio | Google Scholar

Professional Summary

AI/ML Engineer and Data Scientist with 3+ years of experience developing production-grade Machine Learning, Generative AI, Computer Vision, and Retrieval-Augmented Generation (RAG) systems. Skilled in building multimodal AI applications, LLM-powered pipelines, Voice AI systems, and scalable API-driven architectures using Python, FastAPI, and cloud platforms. Experienced in designing intelligent automation solutions involving OCR, NLP, vector search, and real-time AI workflows for healthcare, research, and enterprise applications.

Key Skills

Programming Languages: Python, SQL, R

Data Analysis: Pandas, NumPy, EDA, Feature Engineering

Machine Learning: Scikit-learn, LLM, GenAI, RAG, MCP-Agentic, Hugging Face

Data Visualization: Tableau, Matplotlib, Seaborn

Cloud & Big Data: AWS (S3), PySpark, Docker, Git, MLflow

Web Development & APIs: Django, RESTful APIs, FastAPI, JavaScript

Professional Experience

AI/ML Engineer

Jan 2026 – Present

Idyaite Technologies Pvt. Ltd.

- Developed end-to-end Generative AI and Computer Vision pipelines integrating OCR, embeddings, hybrid retrieval, and LLM-based reasoning for enterprise automation and intelligent data processing.
- Designed and deployed advanced RAG systems using vector databases, dense/sparse retrieval, reranking, and context compression for intelligent document querying and knowledge retrieval.
- Built multimodal RAG solutions capable of interpreting complex 2D hydraulic engineering diagrams and technical documents for visual and textual reasoning.
- Developed multilingual AI voice agents using Twilio, Claude Haiku 3.5, Sarvam AI, and Deepgram with real-time Speech-to-Text → LLM → Text-to-Speech pipelines achieving sub-second latency.
- Implemented cost optimization techniques including dynamic model routing, selective retrieval, context compression, and response caching to reduce inference and API costs in production environments.
- Deployed and optimized Qwen 2.5 (7B) models on AWS GPU infrastructure for scalable and cost-efficient AI inference.
- Built intelligent video frame extraction pipelines using motion detection, blur scoring, and frame quality evaluation, along with automated PDF, DOCX, and Excel comparison systems using Python libraries.
- Designed scalable RESTful AI microservices using FastAPI and implemented face recognition systems for attendance and identity verification using Python computer vision frameworks.

Data Scientist

Nov 2025 – Jan 2026

Indo-Sakura Software Pvt. Ltd. Japan K.K.

- Designed and deployed end-to-end AI/ML and GenAI solutions for multilingual (English–Japanese) enterprise applications, covering data ingestion, modeling, and production deployment.
- Built high-performance RAG systems using optimized embeddings, FAISS vector stores, and precision retrieval to enable accurate document-based question answering.
- Developed MCP-based agentic AI systems for autonomous planning, document intelligence, and workflow automation across business processes.
- Reduced LLM latency and improved inference speed using quantization, batching, and caching techniques for real-time production workloads.
- Implemented OCR and multilingual document text extraction pipelines for structured and unstructured documents such as PDFs and scanned images.
- Collaborated with product and engineering teams to convert business needs into secure, production-ready AI systems.
- Utilized AWS (Lambda, S3) and PySpark for scalable distributed data processing and large-scale model deployment.

Data Scientist, Centre for Integrative Omics Data Science (CIODS)	Feb 2023 – Nov 2025
• Built and deployed ML models using Python, scikit-learn, and Prophet to support forecasting and predictive analytics across structured and unstructured datasets. • Developed GenAI solutions including RAG pipelines and MCP-agentic systems using LLMs, LangChain, and FAISS for intelligent retrieval and planning. • Deployed LLM-based applications via FastAPI, Django, and Hugging Face; implemented CI/CD workflows using GitHub Actions and Docker for scalable model delivery. • Developed financial risk and anomaly detection models leveraging Python & SQL to identify irregular transaction patterns, improving detection accuracy by 15%. • Collaborated with cross-functional teams to translate analytical insights into business strategy and risk-mitigation decisions. • Monitored deployed models for drift, scheduled retraining, and applied version control for reproducibility and governance. • Performed data extraction, feature engineering, and optimization using SQL, Pandas, and NumPy for large-scale datasets.	

Data Science Intern, Knowledge Solutions India (Affiliated with Microsoft)	Feb 2021 – Apr 2021
• Developed machine learning models for data analysis and prediction tasks, improving accuracy by 15%. • Conducted exploratory data analysis (EDA) to uncover trends and patterns, facilitating data-driven decision-making. • Created visualizations using Matplotlib and Seaborn to communicate findings effectively to stakeholders.	

Education

Bachelor of Information Science and Engineering, Visvesvaraya Technological University , CGPA: 8.98	2019 - 2023
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Certifications

Introduction to Artificial Intelligence - IBM
 GenAI Powered Data Analytics Job Simulation - Forage
 SQL (Advanced) – HackerRank
 Introduction to Data Science in Python – University of Michigan
 Tableau Training – Simplilearn

Publications

Positional distribution and conservation of major phosphorylated sites in the human kinome , Frontiers	2025
Elucidating the phosphoregulatory network of predominant phosphosite in AXL kinase: an integrative bioinformatic approach, Springer	2024
A global phosphosite-correlated network map of Thousand And One Kinase 1 , ScienceDirect	2024
Crop Maturity Detection and Automatic Sprinkler Irrigation using IoT and Machine Learning, HBRP	2024
ZMYM3 S464: a potential phospho-regulatory hub in epigenetic remodeling and oncogenesis, Molecular Genetics and Genomics	2026
Dissecting the Phospho-Regulatory Landscape of Protein Kinase N1 (PKN1) and Its Downstream Signaling, International Journal of Molecular Sciences	2026
Proteome-Wide Analysis of Functional Phosphosites in the FGFR Family of Proteins, Proteomes	2026
Phosphorylation-Dependent Regulation and Interactions of the Neutral Amino Acid Transporter SLC6A15, OMICS: A Journal of Integrative Biology	2026
Decoding the Evolution and Co-Phospho-Regulatory Networks of SHROOM Proteins, Journal of Molecular Evolution	2026
Exploring the AAK1 Phospho-signaling Landscape Through Phospho-centric Co-regulation Network, Cell Biochemistry and Biophysics	2026

Google Scholar: 30+ Publications | 100+ Citations